# Syllabus for AAE 55500 – Mechanics of Composite Materials
## On Site and Distance delivery Spring 2026
## 3:30-4:20 pm MWF 1/12/26 – 5/2/26
## 2555 Wang Hall

This course provides students with a background in modern lightweight composite materials which are being used in an ever-increasing range of applications and industries. Basic knowledge of composites will allow engineers to understand the issues associated with using these materials, as well as gain insight into how their usage differs from metals, and ultimately be able to use composites to their fullest potential. Topics covered include: current and potential applications of composite materials, fibers, matrices, manufacturing methods for composites, review of elasticity of anisotropic solids, methods for determining mechanical properties of heterogeneous materials, micromechanics of continuous and discontinuous fiber systems, laminated plate analysis, static analyses of laminated composites, edge effects in laminates and both macroscopic and microscopic failure analysis of composite materials and laminates.

**Topics Covered:**
Fibers, Matrices, and Composite Materials; Elasticity of Anisotropic Materials; Analysis of Lamina; Effective Elastic Moduli; Elastic Analysis of Composite Laminates; Strength Criteria for Composites; De-homogenization for Micro Strength; Interlaminar Fracture; Unusual Bending Characteristics; and Free-edge Effects.

**At the end of this course, student should be able to**:
Describe the characteristics and the manufacturing principles of composite laminates. Carry out thermoelastic analyses for the determination of the stress and strain state in a multi-axial laminate subjected to in-plane and bending resultants. Predict the strength of multi-axial laminates. Carry out interlaminar fracture calculations for unidirectional laminates. Understand the bending-twisting-extensional coupling in symmetric and unsymmetric laminates. Determine the stress state at the free-edge of a finite-width laminate.

**Grading:**
The grade will be made up of three components: homework submissions, two-hourly exams and a final exam. Each of the components will comprise 1/3 of the grade. The final averages required for each grade level are: A (90-100%), B (80-89%), C (70-79%). In place of makeup exams, the points missed will be added to the worth of the final exam. Home work will be due when assigned.

**Lecture Format:**
The lecture format will utilize PowerPoint slides to summarize the content.

**Textbook:**
No textbook will be required, but the notes of Professor C.T. Sun can serve as a reference for a large portion of the course: Mechanics of Composite Materials and Laminates, Lecture Notes for A&AE 555, C. T. Sun, School of Aeronautics and Astronautics, Purdue University, West Lafayette, Indiana, Spring 2001. **The lecture notes will be provided in pdf format to all registered students.**